Optimization of Physics-Informed Neural Networks

In this project, you will investigate the performance of optimization algorithms when training a **Physics-Informed Neural Network (PINN)** to solve a partial differential equation (PDE). PINNs have not been covered in the course. You are therefore expected to learn the basic methodology independently and use it as a framework for studying optimization.

1. Learn & implement a PINN

- Study the basic PINN methodology and consult at least **3 scientific references**.
- Formulate a PDE solution problem as an optimization problem using a neural network.
- Implement a PINN.
- Validate your implementation on a **simple PDE with a known solution**.

2. Compare optimization methods

Train the same PINN using at least GD with Armijo line-search, SGD, SGD with Momentum, Adam, and one additional PyTorch optimizer.

- Investigate relevant hyperparameters (e.g., learning rate).
- Use a fair and clearly justified computational budget.
- Repeat important experiments for at least **3 random initializations**.
- Compare both **optimization performance** and **PDE-solution accuracy**.

Possible quantities to report include loss, PDE residual, solution error, gradient norm, number of iterations, and computational time.

3. Investigate a challenging PDE

Choose one PDE from the following list: *nonlinear Poisson; Burgers; Allen–Cahn; reaction-diffusion; advection–diffusion; high-frequency Helmoltz.* Formulate a **specific research question and hypothesis** concerning the optimization of the PINN. Perform a controlled numerical investigation and explain the observed behaviour using concepts from optimization.

4. Perform one deeper investigation

For the challenging PDE, investigate at least one of the following questions:

- How does the choice of collocation points affect the optimization?
- How do the PDE parameters affect the optimization?
- How does the weighting of the PDE and boundary losses affect the optimization?

5. Analysis & Communication

Your conclusions should address questions such as:

- Which methods perform best, and according to which metric?
- How robust are the methods?
- Does low PINN loss imply an accurate PDE solution?
- How does PDE difficulty affect optimization?

Deliverables

Report: maximum **10 pages**, including: PINN formulation and PDEs; optimization methods; experimental methodology; results; discussion and conclusions; references.

Code: reproducible implementation of the main experiments.

AI log: Document (i) which tools you used and for what, (ii) two cases where AI gave a wrong or misleading answer and how you caught it, (iii) for each AI-generated code block in your submission, the test that validates it, and (iv) one part of the project where AI was unhelpful and why.